INCH-POUND

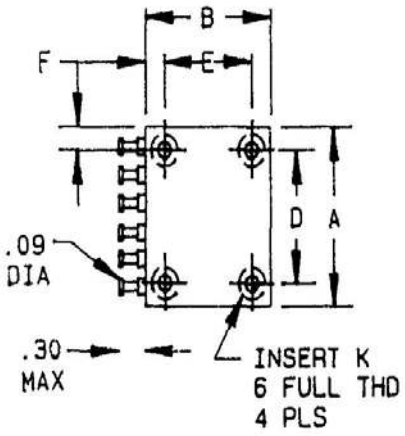
MIL-PRF-27/337E
25 July 2008
SUPERSEDING
MIL-PRF-27/337D
22 January 1992

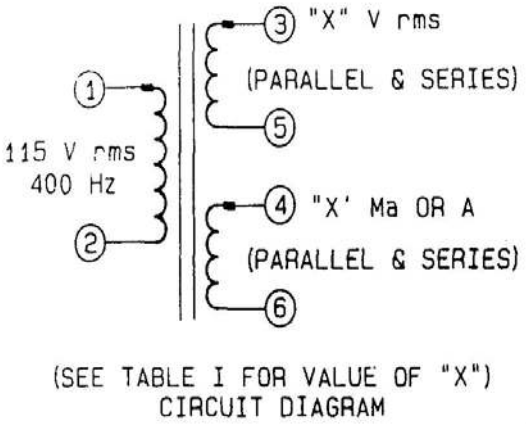
PERFORMANCE SPECIFICATION SHEET

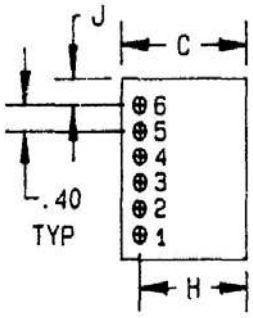
TRANSFORMERS, POWER, 150 AND 200 VOLTAMPERES, 400 HZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of specification MIL-PRF-27.







Inches	mm
.02	0.5
.09	1.3
.30	7.6
.40	10.2

Inches		mm
A	2.81 ±.02	71.4 ±.5
B	2.12 ±.02	53.8 ±.5
C	2.43 ±.02	61.7 ±.5
D	2.21 ±.02	56.1 ±.5
E	1.52 ±.02	38.6 ±.5
F	0.30 ±.04	7.6 ±1.0
H	2.23 ±.04	56.6 ±1.0
J	0.40 ±.04	10.2 ±1.0
K	.164-32 UNC 2B	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information.
3. Marking shall be on top and sides of case.
4. Dimensional tolerance shall be ±.02 unless otherwise specified.
5. For series output, connect terminals 4 and 5 together. For parallel output, connect pins 3 to 4 and 5 to 6.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage (1-2): 115 volts rms, 400 ±20 hertz.

Secondary voltage and current (3-6): See table I.

Voltampere: See table I.

Working voltage (1-2): 535 volts peak.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy fiberglass or diallyl phthalate.

Terminals: Turret type.

Height: .300 inch, maximum.

Weight: 2 pounds, maximum (PIN's -01 through -10) and 2.2 pounds, maximum (PIN's -11 through -19).

Operating temperature range: -55°C to +130°C.

Altitude: 70,000 feet.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage (each winding, secondary windings connected in series):

At sea level: 1500 volts rms.

At reduced barometric pressure: 1.25 times the peak working voltage specified.

Electrical characteristics:

Rated load: With 115 volts rms and 400 Hz in (1-2), and rated current in secondary, the voltage across (3-6) shall be as specified in table I, 5 percent.

Efficiency: 90 percent minimum at rated load.

Temperature rise: 45°C with 115 volts rms, 400 hertz across (1-2) at an ambient temperature of 85°C, full load terminals (3-6), for PIN's -01 through -10.

Temperature rise: 45°C with 115 volts rms, 400 hertz across (1-2) at an ambient temperature of 70°C, full load terminals (3-6), for PIN's -11 through -19.

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Regulation: $\frac{Voltage\ (no\ load) - Voltage\ (rated\ load)}{Voltage\ (rated\ load)}\ x\ 100$

Shall not exceed 10 percent.

Polarity: Additive, with terminals 2 and 3, and 5 and 4 connected.

Marking location: See figure 1.

Part or Identifying Number (PIN). M27/337-(dash number from table I).

VERIFICATION:

Extent of qualification:

Qualification testing and approval to M27/332-10 and M27/337-19 shall be sufficient to grant qualification approval to MIL-PRF-27/332 through MIL-PRF-27/337 inclusive, all parts.

Qualification testing and approval to M27/337-19 shall be sufficient to grant qualification approval to M27/337-01 through M27/337-19.

TABLE I. Electrical ratings.

Dash number M27/337	Secondary voltage (3-6) (V rms) ±5%		Secondary current (3-6) (amperes)		Volt-ampere rating
	Series (CT)	Parallel	Series	Parallel	
-01	10	5	15	30	150
-02	12.6	6.3	11.9	23.8	150
-03	16	8	9.37	18.7	150
-04	20	10	7.50	15	150
-05	24	12	6.25	12.5	150
-06	34	17	4.41	8.82	150
-07	40	20	3.75	7.50	150
-08	56	28	2.68	5.36	150
-09	88	44	1.70	3.40	150
-10	120	60	1.25	2.50	150
-11	12.6	6.3	15.85	31.70	200
-12	16	8	12.50	25	200
-13	20	10	10	20	200
-14	24	12	8.35	16.70	200
-15	34	17	5.90	11.80	200
-16	40	20	5	10	200
-17	56	28	3.57	7.14	200
-18	88	44	2.27	4.54	200
-19	120	60	1.66	3.53	200

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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents.

MIL-PRF-27
MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2008-052)

Review activities:
Army – AR, CR4
Navy - AS, MC, OS
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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